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ALY 6050

Module 2 Assignment

Benefit-Cost Analysis

Introduction

The aim of this report is to look at how a company assesses the benefit-cost ratios of two dams to determine which one should be selected for an upcoming project. The benefit-cost analysis that the corporation will use entails simulations of both dams that will lead to creating graphical and tabular frequency distributions. Other calculations of the values will help provide evidence as to which dam is better suited for future project considerations. Several concepts that were learned from this week's module were incorporated for completion of this assignment.

Analysis

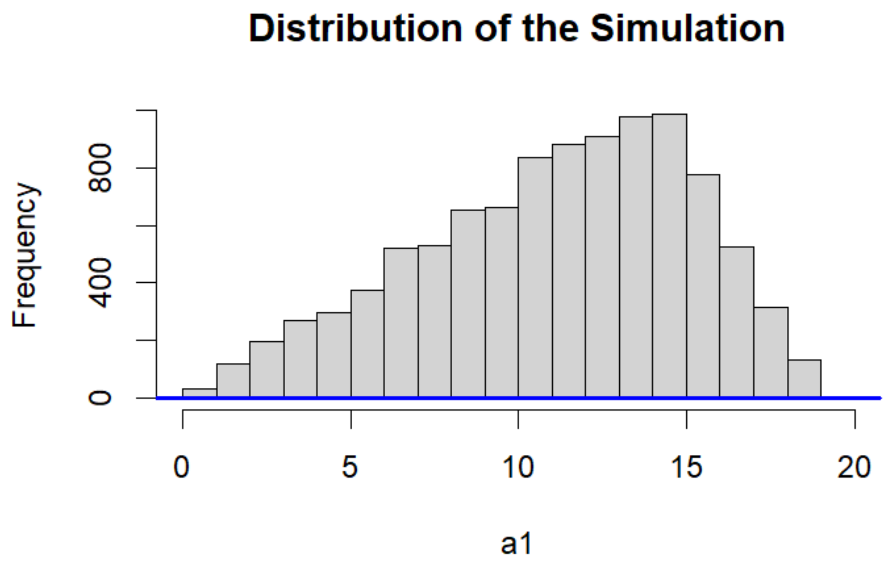
Part 1

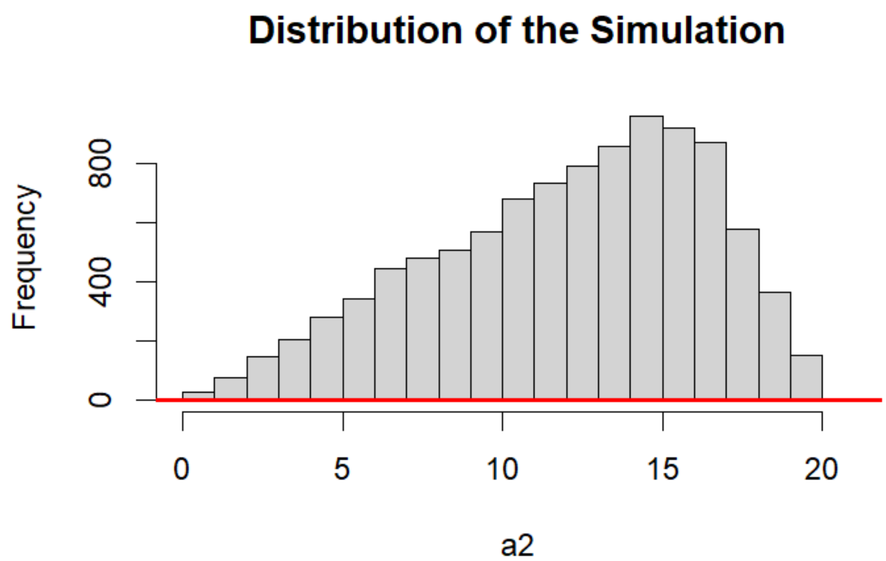
i. Perform a simulation of 10,000 benefit-cost ratios for Dam #1 project and 10,000 such simulations for Dam #2 project. Note that the two simulations should be independent of each other. Let these two ratios be denoted by a1 and a2 for the dams 1 and 2 projects respectively.

```
#(i)Perform Simulation of 10,000 benefit-cost ratios for Dam #1

#Specify the specification of the triangular distribution:
a <-0
b <-19.1
c <-14.2

#Generate ten-million random values according to the standard uniform distribution:
set.seed(123)
N <-10^4
r1 <-runif(N)
head(r1)

#Implement formula to generate ten million triangular random values (labeled as a1):
A <-a+sqrt((b-a)*(c-a)*r1)
B <-b-sqrt((b-a)*(b-c)*(1-r1))
C <-(c-a)/(b-a)
a1 <-ifelse(r1<C,A,B)
```

```
#Perform Simulation of 10,000 benefit-cost ratios for Dam #2

#Specify the specification of the triangular distribution:
d <-0
e <-20.1
f <-15.8

#Generate ten-million random values according to the standard uniform distribution:
set.seed(456)
N <-10^4
r2 <-runif(N)
head(r2)

#Implement formula to generate ten million triangular random values (labeled as a2):
D <-d+sqrt((e-d)*(f-d)*r2)
E <-e-sqrt((e-d)*(e-f)*(1-r2))
F <-(f-d)/(e-d)
a2 <-ifelse(r2<F,D,E)
```

ii. Construct both a tabular and a graphical frequency distribution for a1 and a2 separately (a tabular and a graphical distribution for a1, and a tabular and a graphical distribution for a2- a total of 4 distributions). In your report, include only the graphical distributions and comment on the shape of each distribution.





It seems that both graphs from the simulations of the two dams show a more negatively skewed distribution. A negatively skewed histogram, as observed in random number generation, is characterized by a leftward elongation of the distribution's tail. In such a distribution, the mode tends to be higher than the median, and the median is higher than the mean. Moreover, most data points are clustered on the right side of the distribution, indicating a concentration of values towards higher magnitudes, while a few extreme values contribute to the extension of the left tail. This skewness pattern implies that, in the context of random number generation, the likelihood of obtaining values on the higher end of the range is greater, while occurrences of lower values are comparatively less frequent. The specific shape of this skewed distribution is influenced by the probability distribution underlying the random number generation process.

iii. For each of the two dam projects, perform the necessary calculations in order to complete the following table. Excel users should create the table in Excel with all cells being occupied by the appropriate formulas, and R users should display the table as a "data frame". Remember to create two such tables – one table for Dam #1 and another table for Dam #2. Include both tables in your report.

```
> # Print the data frame
> print(my_table)
                          Dam_1 Observed Theoretical
1     Mean of the Total Benefits     8.12        8.13
2       SD of the Total Benefits     3.04        3.04
3         Mean of the Total Cost    12.30       12.27
4           SD of the Total Cost     3.22        3.26
5 Mean of the Benefit-cost Ratio    11.07        8.60
6   SD of the Benefit-cost Ratio     4.03        3.05
>
```

```
> # Print the data frame
> print(my_table1)
                          Dam_1 Observed Theoretical
1     Mean of the Total Benefits     8.53        8.60
2       SD of the Total Benefits     3.11        3.05
3         Mean of the Total Cost    13.31       13.23
4           SD of the Total Cost     3.41        3.45
5 Mean of the Benefit-cost Ratio    12.01       12.27
6   SD of the Benefit-cost Ratio     4.28        3.26
```

<u>Part 2:</u> Use your observation in Question (ii) of Part 1 to select a theoretical probability distribution that, in your judgement, is a good fit for the distribution of a1. Next, use the Chi-squared Goodness-of-fit test to verify whether your selected distribution was a good fit for the distribution of a1. Describe the rational for your choice of the probability distribution and a description of the outcomes of your Chi-squared test in your report. In particular, indicate the values of the Chi-squared test statistic and the P-value of your test in your report, and interpret those values.

I decided to use the triangular probability distribution as my theoretical probability distribution because it serves as a straightforward continuous model employed in scenarios with restricted sample data and limited population information. This model relies on the available knowledge of a minimum value, a maximum value, and a mode value positioned between these two extremes (Module 2, Lab 2).

```
#Part 2: Create Goodness of Fit Test

#State the hypotheses
#H0: There is no difference in the estimates between the benefits and costs.
#H1: There is a difference in the estimates between the benefits and costs.

#Find the Critical Value
#df = 5, 18, a = 0.05, Critical Value = 4.96

#Set the significant level
alpha <- 0.05
```

```
> #Combine data frames
> dams <- rbind(dam3, dam4)
> dams
  BenefitCost Minimum Mode Maximum
1          b1     1.1  2.0     2.8
2          b2     8.0 12.0    14.9
3          b3     1.4  1.4     2.2
4          b4     6.5  9.8    14.6
5          b5     1.7  2.4     3.6
6          b6     0.0  1.6     2.4
7          c1    13.2 14.2    19.1
8          c2     3.5  4.9     7.4
```

```
> summary(data)
            Df Sum Sq Mean Sq F value Pr(>F)
Costs        1  17.68  17.682   1.946  0.236
Residuals    4  36.35   9.087
```

The Chi-Squared Goodness of Fit test was demonstrated just for the first dam to see if there is a strong relationship between the benefits and costs of the project. With the calculated test value being 1.946, it is less than the critical value of 4.96 which means we should not reject the null hypothesis. There is not enough evidence to conclude that a difference in estimate amounts exist between the two categories that were being compared, with the values being similar to all three columns of benefits and costs in the triangular distribution. The p-value of 0.236 is also very high which also justifies that the null hypothesis should not be rejected.

Part 3

i. Use the results of your simulations and perform the necessary calculations in order to complete the table below. Excel users should create the table in Excel with all cells being occupied by the appropriate formulas, and R users should display the table as a "data frame". Include the completed table in your report.

```
> # Create a data frame
> my_table2 <- data.frame(
+   Dams = dam_titles1,
+   a1 = observed_values1,
+   a2 = theoretical_values1
+ )
> # Print the data frame
> print(my_table2)
                  Dams     a1     a2
1              Minimum  0.130  0.300
2              Maximum 19.030 20.040
3                 Mean 11.070 12.010
4               Median 11.580 12.650
5             Variance 16.220 18.280
6   Standard Deviation  4.030  4.280
7             SKEWNESS -0.420 -0.450
8              P(a1>2)  0.985  0.987
9            P(a1>1.8)  0.988  0.989
10           P(a1>1.5)  0.992  0.993
11           P(a1>1.2)  0.995  0.996
12             P(a1>1)  0.996  0.997
```

ii. In your report, use your observations of the results obtained in parts 1-3 to recommend one of two projects to the management. Explain all your rationales for the project that you have recommended.

From the three parts of the assignment that compares the two dams, they are both pretty similar in the calculations of the benefit-cost analysis. One aspect of the analysis that proves to be consistent in all the tests performed is that the second dam has slightly higher values than the first dam. This could potentially stem from the fact that when a triangular distribution is conducted, it initially showed that the benefits and costs of each factor are higher overall in the second dam.

When looking at the goodness of fit testing, the observed values closely match the theoretical values in statistical analysis or hypothesis testing. This indicates a strong connection between the collected data and the expected values derived from the theoretical model. The alignment also suggests that there is minimal divergence of the observed data from the expected outcomes posited by the assumed theoretical distribution or hypothesis. The final calculations that were tallied in part three are also indicative of the strength of the second damn with slightly better probabilities that are greater than the values of the first dam. The only category that favored the first dam in these calculations was on skewness, but because both are negative it still doesn't mean that the first dam is close to having a normal distribution overall.

Conclusion

A final t-test was performed to see if the probability of the first dam is greater than the second dam. The results show a low t-score, a very high p-value, and a larger mean from the second dam, resulting in a zero probability of the first dam being greater. My recommendation will therefore be that the second dam be selected for the upcoming project based on the better benefit-cost ratios. The benefits of the benefit-cost ratio outweigh the costs and the likelihood of selecting a project like the second dam increases with a higher benefit-cost ratio compared to projects with lower ratios.

```
> # Perform a one-sided t-test
> result <- t.test(a1, a2, alternative = "greater")
> # Extract the p-value
> p_value <- result$p.value
> # Calculate the estimated probability that mean(a1) > mean(a2)
> estimated_probability <- 1 - p_value
> # Print the result
> print(result)

        Welch Two Sample t-test

data:  a1 and a2
t = -16.075, df = 19927, p-value = 1
alternative hypothesis: true difference in means is greater than 0
95 percent confidence interval:
 -1.040892       Inf
sample estimates:
mean of x mean of y
 11.06983  12.01410

> cat("Estimated probability that mean(a1) > mean(a2):", estimated_probability, "\n")
Estimated probability that mean(a1) > mean(a2): 0
```

References

Behboudi, R. (2024). *Lecture - Triangular Probability Distribution.* Northeastern University – ALY 6050.

Bluman, A. G. (2017). Elementary Statistics: A Step-by-Step Approach. 10th edition. McGraw-Hill Education.

ChatGPT. (2024, February 28th). Default (GPT 3.5). https://chat.openai.com/

Evans, J. (2012). Statistics, Data Analysis, and Decision Modeling, 5th Edition. Pearson Publishing.

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